RWANDA POLYTECHNIC

HUYE COLLEGE

**MODULE NAME: DATA MINING AND DATA WAREHOUSING.**

**MODULE CODE : ITLDM801.**

**DEPARTMENT: ICT**

**OPTION: IT**

**LEVEL: 8 YEAR 4**

**CLASS: IT Btech**

# LIBRARY USAGE DATA WAREHOUSE SYSTEM

***A Comprehensive Data Integration Solution for Modern Library Management***


## GROUP 1: MEMBERS

| NAMES | REG NO | MARKS |
|---|---|---|
| Gertrude IRIMASO | 25RP19175 | |
| Cyprien NZAYISENGA | 25RP20156 | |
| Aline AKIMANA | 25RP20978 | |


*Date: 06th Feb 2026*

# TABLE OF CONTENTS


**LIST OF TABLES** ........ 2

**LIST OF FIGURES** ........ 2

**1. Introduction** ........ 3

**1.1 The project successfully delivers:** ........ 3

**2. Problem Statement** ........ 4

**3. Project Overview & Objectives** ........ 4

**3.1 Business Objectives:** ........ 4

**3.2 Technical Objectives:** ........ 4

**4. System Architecture** ........ 4

**4.1 Components:** ........ 5

**5. Data Warehouse Schema Design** ........ 5

**5.1 Entity Relationship Diagram (ERD)** ........ 5

**5.2 Dimensional Model Justification** ........ 7

**6. ETL Process Design & Implementation** ........ 7

**6.1 ETL Workflow** ........ 7

**6.2 NULL Handling Strategy** ........ 7

**6.3 Data Quality Framework** ........ 8

**7. Technology Stack** ........ 8

**8. Implementation Details** ........ 9

**8.1 Database Scripts** ........ 9

**8.2 ETL Scripts** ........ 9

**8.3 Data Validation & Cleansing** ........ 9

**9. OLAP & Analytics Capabilities** ........ 9

**9.1 Sample Business Queries Answered:** ........ 12

**10. Security, Compliance & Access Control** ........ 12

**11. Analysis and Reporting** ........ 14

**12. Performance Considerations** ........ 16

**13. Challenges & Solutions** ........ 16
**14. Future Enhancements** ........ 17
**15. Conclusion** ........ 17

## LIST OF TABLES

**Table 1 : Showing NULL Handling Strategy** ........ 8
**Table 2 : Showing Technologies used** ........ 8
**Table 3 : Showing Challenges & Solutions we faced** ........ 17

## LIST OF FIGURES

**Figure 1 : Showing Three-Tier Architecture** ........ 5
**Figure 2 : Showing Entity Relationship Diagram** ........ 6
**Figure 3 : Showing ETL Process Flow** ........ 7
**Figure 4 : Showing DRILL-DOWN Analysis (Year to Month)** ........ 10
**Figure 5 : Showing DRILL-DOWN Analysis (Year to quarter)** ........ 10
**Figure 6 : Showing Drill down to daily: January 2024** ........ 10
**Figure 7 : Showing Roll-up: Daily → Weekly → Monthly summaries** ........ 11
**Figure 8 : Showing Slicing: Filter by time period (e.g., Q1 2024)** ........ 11
**Figure 9 : Showing Dicing For Engineering students in March** ........ 12
**Figure 10 : Shows access based on user roles** ........ 13
**Figure 11 : Showing masked the student's IDs** ........ 13
**Figure 12 : Showing Executive Dashboard** ........ 14
**Figure 13 : Showing Operational Dashboard** ........ 15
**Figure 14 : Showing Department Dashboard** ........ 16

# 1. Introduction

This report details the design and implementation of **a Library Usage Data Warehouse** that consolidates data from three heterogeneous sources : **book lending, digital resource downloads, and study room bookings** : into a unified, analytics-ready platform. The system replaces manual, error-prone reporting with an automated, scalable, and secure data warehouse solution that enables the University Library to perform advanced OLAP operations, generate interactive dashboards, and answer critical business questions in near real-time.

## 1.1 The project successfully delivers:

- **A three-tier architecture** with robust ETL/ELT processes
- **A star schema** dimensional model optimized for analytics
- Advanced **NULL handling** and data quality frameworks
- **Role-based access control** and security measures
- Interactive **dashboards** for library management
- Full documentation and scalability planning

## 2. Problem Statement

The University Library currently operates with three independent legacy systems, leading to inconsistent reporting, data silos, and inefficiencies. This project addresses these challenges by designing and implementing a centralized data warehouse that integrates, cleanses, and transforms data into actionable insights. The solution follows industry best practices in data warehousing, dimensional modeling, and ETL design.

## 3. Project Overview & Objectives

### 3.1 Business Objectives:

- Reduce manual reporting time from 15+ hours/month to automated processes
- Enable self-service analytics for library staff and committee
- Provide historical trend analysis and predictive insights
- Ensure data consistency and accuracy across all systems

### 3.2 Technical Objectives:

- Consolidate data from MySQL, Excel, and CSV sources
- Implement a scalable dimensional model (star schema)
- Develop a robust ETL pipeline with data quality controls
- Enable OLAP operations (drill-down, roll-up, slicing, dicing)
- Deliver interactive visualizations and dashboards

## 4. System Architecture

The system follows a **three-tier architecture** as illustrated **in Figure 1**:

**Data Source Layer → Staging Layer → ETL/ELT Layer → Data Warehouse Layer → Presentation Layer**

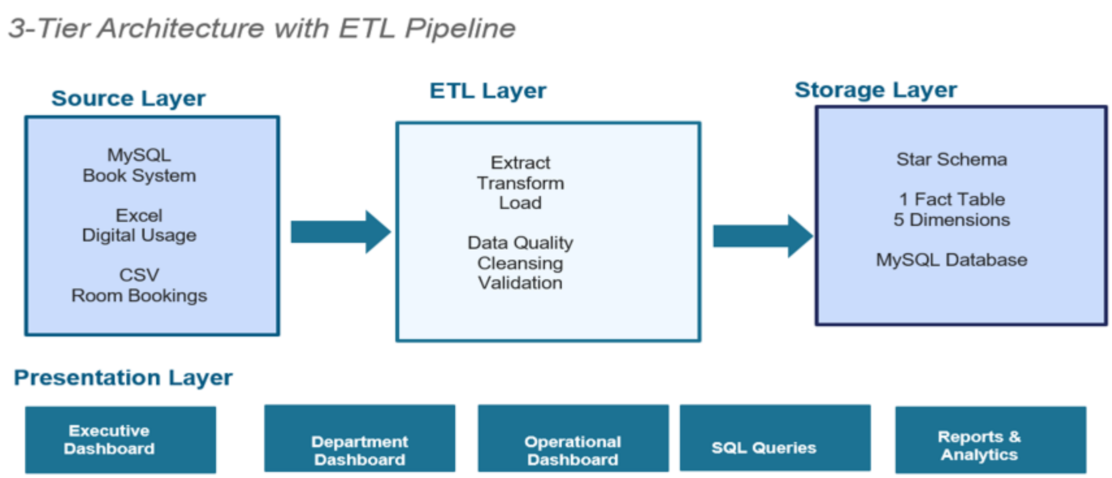



**Figure 1 : Showing Three-Tier Architecture**

## 4.1 Components:

- Data Source Layer: CSV, Excel, MySQL
- Staging Layer: Temporary tables for raw data
- ETL/ELT Layer: Python/Pandas-based transformation
- Data Warehouse Layer: MySQL with star schema
- Presentation Layer: Dashboards (Power BI)

# 5. Data Warehouse Schema Design

## 5.1 Entity Relationship Diagram (ERD)

The warehouse uses a **star schema** with one central fact table (fact_library_usage) linked to multiple dimension tables. **See Figure 2** for the complete ERD.

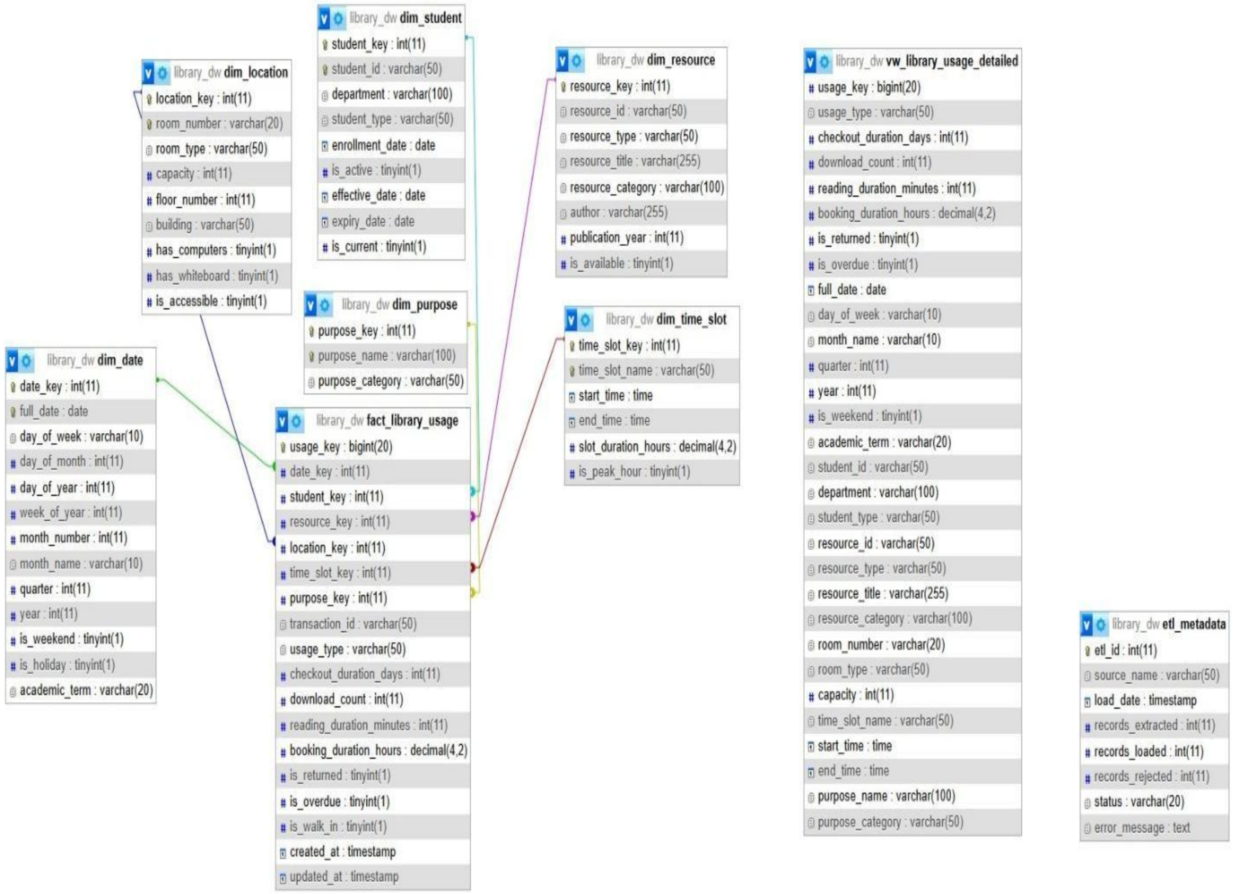



**Figure 2 : Showing Entity Relationship Diagram**

**Dimension Tables:**

- dim_date
- dim_student
- dim_resource
- dim_location
- dim_time_slot
- dim_purpose

**Fact Table:**

- fact_library_usage (measures: checkout duration, download count, booking hours)

## 5.2 Dimensional Model Justification

- **Star schema** chosen for query performance and simplicity
- Supports business-friendly reporting
- Enables fast joins and OLAP operations
- Scalable for future data marts

# 6. ETL Process Design & Implementation

## 6.1 ETL Workflow

The ETL process follows the flow shown **in Figure 3**:

*Extract, Transform, Load Process*


**Figure 3 : Showing ETL Process Flow**

**1. Extract:** Load data from source files into staging tables

**2. Transform:** Cleanse, standardize, and handle NULLs

**3. Load:** Populate dimension and fact tables

## 6.2 NULL Handling Strategy

**NULLs are handled contextually**: Missing or incomplete data is addressed based on the field and analysis context. For example, numerical fields may be filled with averages, categorical fields with default values, or ignored in specific calculations to maintain data integrity.

NULLs are handled contextually: in **Table 1 below** .

| Field | Strategy | Action |
|---|---|---|
| ReturnDate | Flag as "Not Returned" | Preserve NULL, add flag |
| Duration_Minutes | Impute with resource-type average | Replace NULL with calculated average |
| StudentID (room) | Flag as "Walk-in" | Replace NULL with "UNKNOWN", add flag |
| Department | Impute from other records | Lookup from student history |

**Table 1 : Showing NULL Handling Strategy**

## 6.3 Data Quality Framework

- Automated checks ensure:
- Completeness: Missing values tracked
- Consistency: Standardized naming (e.g., "CS" → "Computer Science")
- Accuracy: Date formats unified to ISO 8601
- Uniqueness: Duplicate records removed

## 7. Technology Stack

**Technology Stack** outlines the key tools and technologies used in the system. The **Database** stores and manages data, **ETL** handles extraction, transformation, and loading of data, **Visualization** tools display insights via dashboards, **Version Control** manages code changes, and **Documentation** ensures proper reporting and record-keeping. **Below table 2** summarizes it .

| Component | Tool/Technology |
|---|---|
| Database | MySQL |
| ETL | Python 3.9 + Pandas |
| Visualization | Power BI / Tableau |
| Version Control | Git |
| Documentation | MS Word , PDF |

**Table 2 : Showing Technologies used**

## 8. Implementation Details

### 8.1 Database Scripts

- 01_create_database.sql : Database and schema creation
- 02_create_staging_tables.sql : Staging table DDL
- 03_create_dimension_tables.sql : Dimension tables
- 04_create_fact_tables.sql : Fact table with indexes
- 05_create_indexes.sql : Performance optimization

### 8.2 ETL Scripts

**etl_library_dw.py**– Main ETL pipeline with NULL handling

Logs: **etl_process.log**, **data_quality_report.txt**

### 8.3 Data Validation & Cleansing

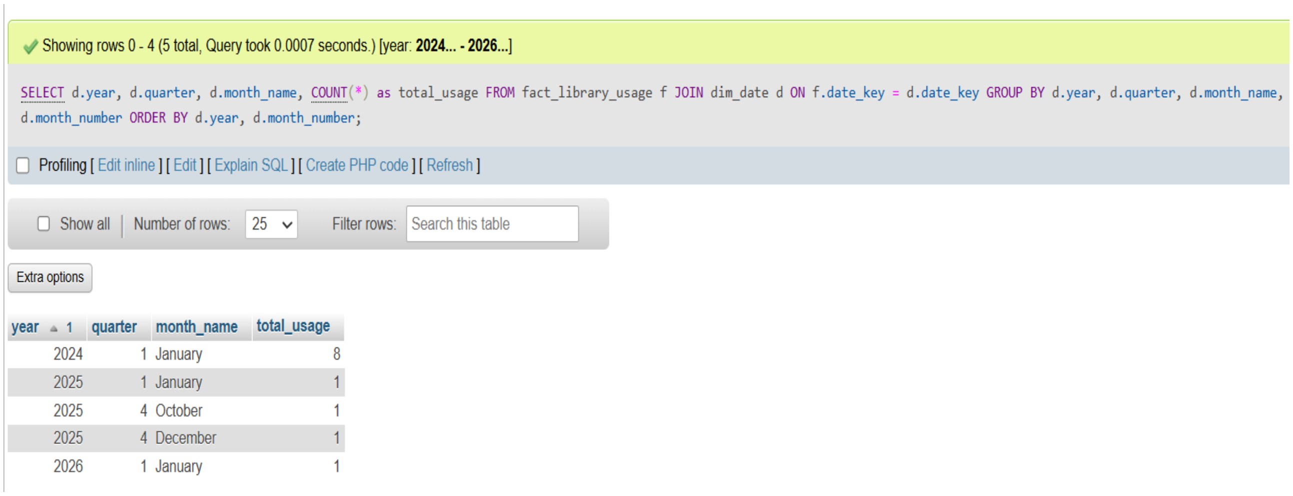
- Standardized department names
- Unified date formats
- Removed duplicates using transaction IDs
- Imputed missing values with business logic

## 9. OLAP & Analytics Capabilities

**The warehouse supports:**

1. Drill-down analysis is a technique used to analyze data by moving from a summarized level to more detailed levels. In time-based analysis, data is first viewed at the **Year** level to understand overall performance. The analysis can then be drilled down to the **Quarter** level to identify variations within the year. Further drill-down to the **Month** level provides more detailed insights, allowing users to observe monthly trends and patterns. This approach helps decision-makers understand performance changes over time in a clear and structured manner. **See In Figure 4,5, 6** Drill-down: Year → Quarter → Month → Day

.



Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.) [year: **2024...** - **2026...**]

```
SELECT d.year, d.quarter, d.month_name, COUNT(*) as total_usage FROM fact_library_usage f JOIN dim_date d ON f.date_key = d.date_key GROUP BY d.year, d.quarter, d.month_name, d.month_number ORDER BY d.year, d.month_number;
```

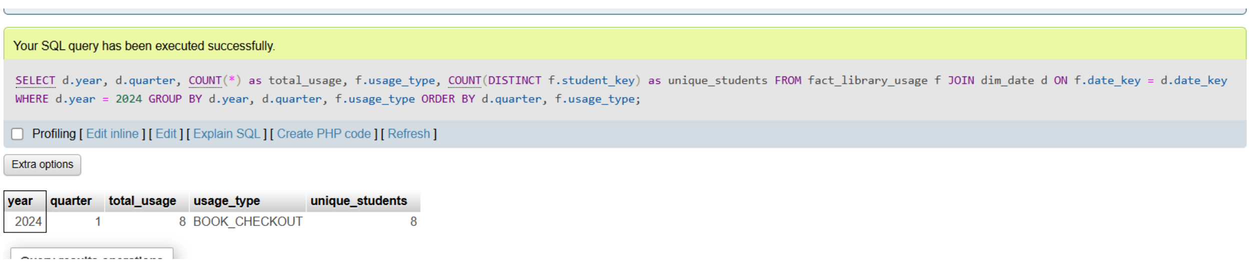
Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| year 1 | quarter | month_name | total_usage |
|---|---|---|---|
| 2024 | 1 | January | 8 |
| 2025 | 1 | January | 1 |
| 2025 | 4 | October | 1 |
| 2025 | 4 | December | 1 |
| 2026 | 1 | January | 1 |

**Figure 4 : Showing DRILL-DOWN Analysis (Year to Month)**



Your SQL query has been executed successfully.

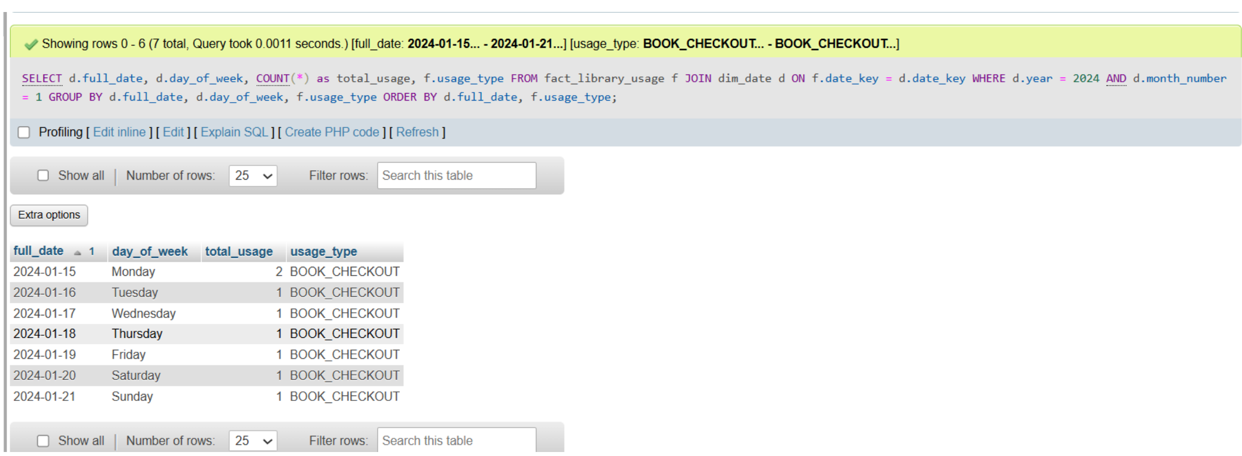
```
SELECT d.year, d.quarter, COUNT(*) as total_usage, f.usage_type, COUNT(DISTINCT f.student_key) as unique_students FROM fact_library_usage f JOIN dim_date d ON f.date_key = d.date_key WHERE d.year = 2024 GROUP BY d.year, d.quarter, f.usage_type ORDER BY d.quarter, f.usage_type;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Extra options

| year | quarter | total_usage | usage_type | unique_students |
|---|---|---|---|---|
| 2024 | 1 | 8 | BOOK_CHECKOUT | 8 |

**Figure 5 : Showing DRILL-DOWN Analysis (Year to quarter)**



Showing rows 0 - 6 (7 total, Query took 0.0011 seconds.) [full_date: **2024-01-15...** - **2024-01-21...**] [usage_type: **BOOK_CHECKOUT...** - **BOOK_CHECKOUT...**]

```
SELECT d.full_date, d.day_of_week, COUNT(*) as total_usage, f.usage_type FROM fact_library_usage f JOIN dim_date d ON f.date_key = d.date_key WHERE d.year = 2024 AND d.month_number = 1 GROUP BY d.full_date, d.day_of_week, f.usage_type ORDER BY d.full_date, f.usage_type;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

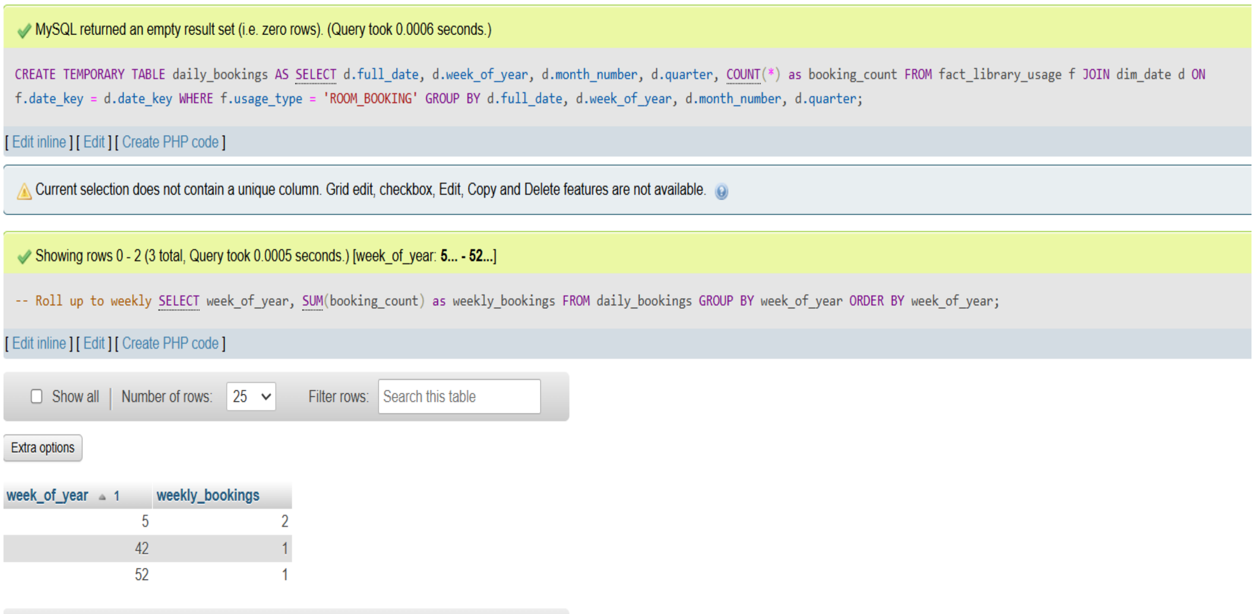
| full_date 1 | day_of_week | total_usage | usage_type |
|---|---|---|---|
| 2024-01-15 | Monday | 2 | BOOK_CHECKOUT |
| 2024-01-16 | Tuesday | 1 | BOOK_CHECKOUT |
| 2024-01-17 | Wednesday | 1 | BOOK_CHECKOUT |
| 2024-01-18 | Thursday | 1 | BOOK_CHECKOUT |
| 2024-01-19 | Friday | 1 | BOOK_CHECKOUT |
| 2024-01-20 | Saturday | 1 | BOOK_CHECKOUT |
| 2024-01-21 | Sunday | 1 | BOOK_CHECKOUT |

Show all | Number of rows: 25 Filter rows: Search this table

**Figure 6 : Showing Drill down to daily: January 2024**

2. **Roll-Up Analysis (Day → Week → Month → Quarter)** is a data aggregation technique that summarizes detailed data into higher-level views. For example, daily

data can be rolled up into weekly, monthly, and quarterly totals to provide broader trends and overall performance insights. It is shown **in the figure 7** below .


MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```sql
CREATE TEMPORARY TABLE daily_bookings AS SELECT d.full_date, d.week_of_year, d.month_number, d.quarter, COUNT(*) as booking_count FROM fact_library_usage f JOIN dim_date d ON f.date_key = d.date_key WHERE f.usage_type = 'ROOM_BOOKING' GROUP BY d.full_date, d.week_of_year, d.month_number, d.quarter;
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.) [week_of_year: **5... - 52...**]

```sql
-- Roll up to weekly SELECT week_of_year, SUM(booking_count) as weekly_bookings FROM daily_bookings GROUP BY week_of_year ORDER BY week_of_year;
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

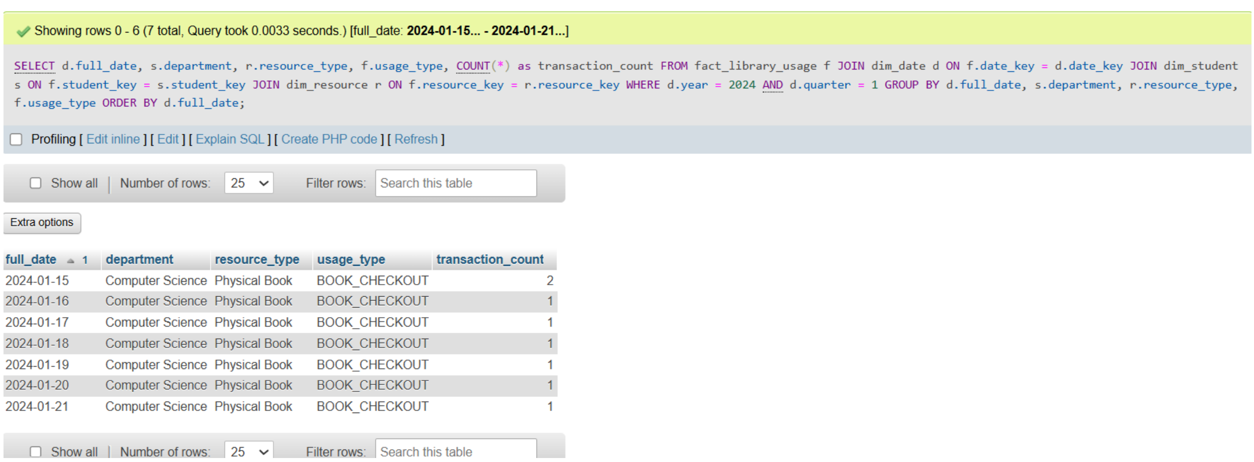
Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| week_of_year | weekly_bookings |
|---|---|
| 5 | 2 |
| 42 | 1 |
| 52 | 1 |

**Figure 7 : Showing Roll-up: Daily → Weekly → Monthly summaries**

3. **Slicing** focuses on analyzing data for a **specific time period** or subset, allowing users to examine trends or performance within that defined segment. See in the **figure 8 below** .


Showing rows 0 - 6 (7 total, Query took 0.0033 seconds.) [full_date: **2024-01-15... - 2024-01-21...**]

```sql
SELECT d.full_date, s.department, r.resource_type, f.usage_type, COUNT(*) as transaction_count FROM fact_library_usage f JOIN dim_date d ON f.date_key = d.date_key JOIN dim_student s ON f.student_key = s.student_key JOIN dim_resource r ON f.resource_key = r.resource_key WHERE d.year = 2024 AND d.quarter = 1 GROUP BY d.full_date, s.department, r.resource_type, f.usage_type ORDER BY d.full_date;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| full_date | department | resource_type | usage_type | transaction_count |
|---|---|---|---|---|
| 2024-01-15 | Computer Science | Physical Book | BOOK_CHECKOUT | 2 |
| 2024-01-16 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |
| 2024-01-17 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |
| 2024-01-18 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |
| 2024-01-19 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |
| 2024-01-20 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |
| 2024-01-21 | Computer Science | Physical Book | BOOK_CHECKOUT | 1 |

Show all | Number of rows: 25 Filter rows: Search this table

**Figure 8 : Showing  Slicing: Filter by time period (e.g., Q1 2024)**

4. **Dicing** is a multi-dimensional analysis technique that allows viewing data from different perspectives, such as examining **students in different departments in January**. It helps focus on specific slices of data for detailed insights. As Shown in the **figure 9 below** .


**Figure 9 : Showing Dicing For Engineering students in March**

### 9.1 Sample Business Queries Answered:

1. Monthly usage trends (physical vs. digital)
2. Peak study room times
3. Top 10 borrowed books
4. Average loan duration by department

## 10. Security, Compliance & Access Control

- Role-Based Access Control (RBAC): Admin: Full access , Analyst: Read-only , Department Head: Restricted to department data .

**Role-Based Access Control (RBAC)** manages system access based on user roles. **Admins** have full access, **Analysts** have read-only access, and **Department Heads** can only access data related to their departments. This ensures data security and proper use of the system. **Figure 10** . Shows the different users who have granted access for the system .

```
MariaDB [(none)]> use library_dw;
Database changed
MariaDB [library_dw]> SELECT User, Host FROM mysql.user;
+-------------+-----------+
| User        | Host      |
+-------------+-----------+
| root        | 127.0.0.1 |
| root        | ::1       |
| admin       | localhost |
| analyst     | localhost |
| backup_user | localhost |
| department  | localhost |
| etl         | localhost |
| pma         | localhost |
| root        | localhost |
```

**Figure 10 : Shows access based on user roles**

Data Masking: Student IDs anonymized in reports , Encryption: Data at rest and in transit .

**Data Masking** protects privacy by anonymizing student IDs in reports, while **Encryption** secures data both at rest and during transmission. These measures ensure data confidentiality and prevent unauthorized access. **Figure 11** . Shows how we masked the student's IDs as their sensitive data .

| ←T→ | | | | masked_student_id | department | student_type | enrollment_date |
|---|---|---|---|---|---|---|---|
| ☐ | Edit | Copy | Delete | UNKNOWN-XXXX-UNKNOWN | Unknown | Unknown | 1900-01-01 |
| ☐ | Edit | Copy | Delete | STU-XXXX-001 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-002 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-003 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-004 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-005 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-006 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-007 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-008 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-009 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-010 | Computer Science | Student | 2026-01-29 |
| ☐ | Edit | Copy | Delete | STU-XXXX-012 | Eng | Student | 2026-02-02 |
| ☐ | Edit | Copy | Delete | STU-XXXX-013 | Engineering | Student | 2026-02-02 |

☐ Check all With selected: Edit Copy Delete Export

**Figure 11 : Showing masked the student's IDs**

## 11. Analysis and Reporting

Showing the analysis of data warehouse and the reports using power BI dashboards , that represents how non-technical users can view the reported data, usage trends , peak usage times, etc. .

1. **Executive Dashboard** provides a high-level visual overview of key performance indicators (KPIs), trends, and summaries, enabling executives to make quick, informed decisions. As shown in the **figure 12** below .


**Figure 12 : Showing Executive Dashboard**

2. **Operational Dashboard** displays real-time data, detailed metrics, and day-to-day operational insights, helping managers monitor activities and take immediate action. As shown in below **figure 13** .

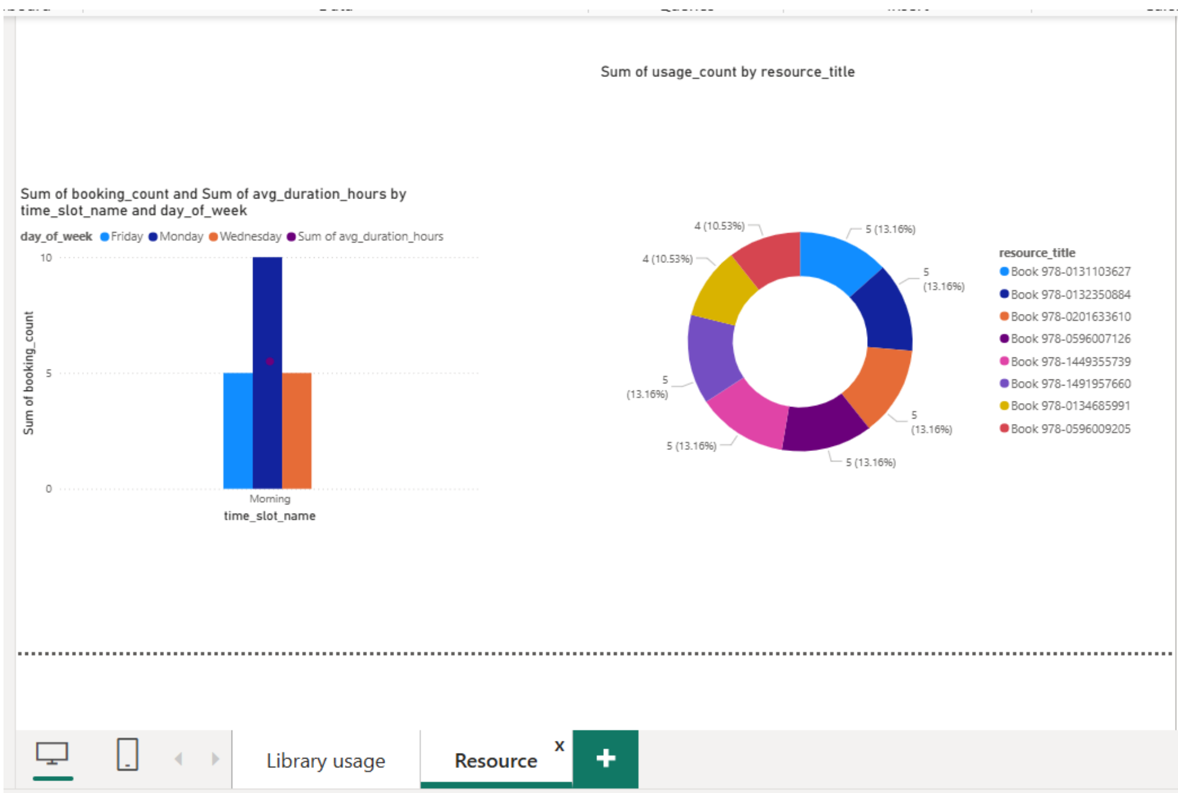



**Figure 13 : Showing Operational Dashboard**

3. **Department Dashboard** provides department-specific insights, metrics, and reports, allowing department heads to monitor performance and make informed decisions within their area. As shown in the **below figure 14 .**

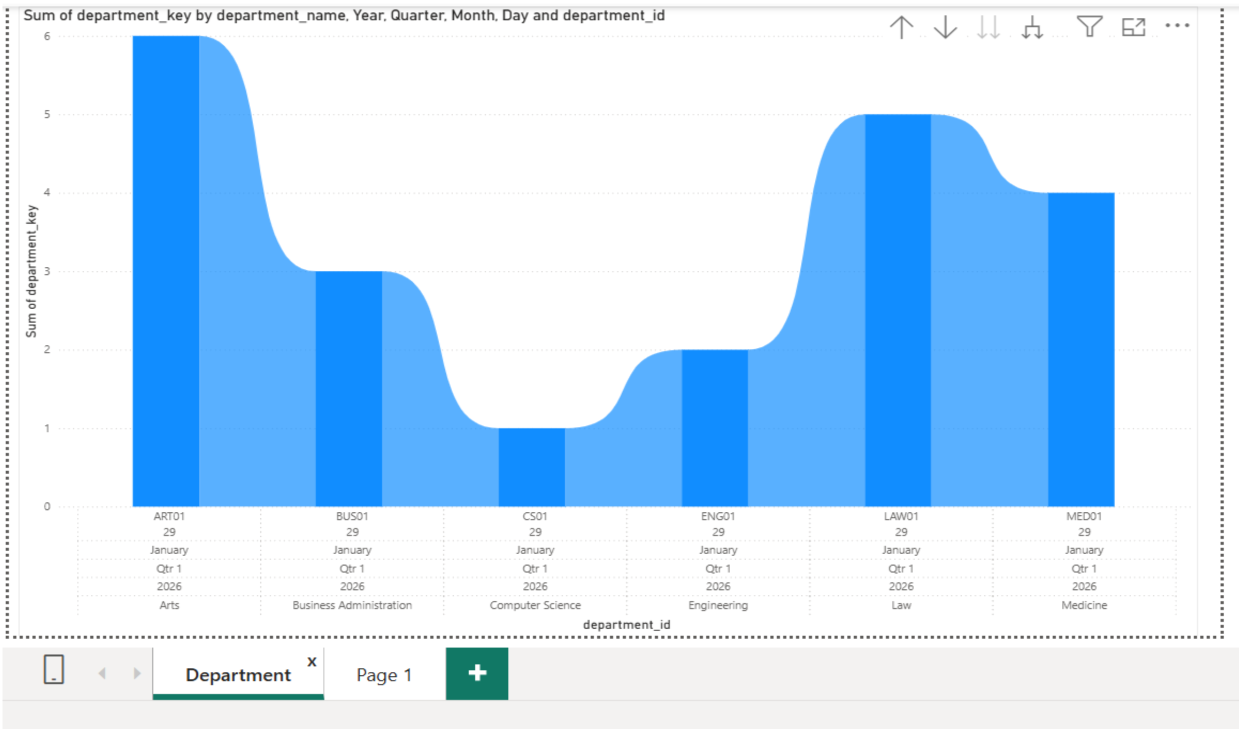



**Figure 14 : Showing Department Dashboard**

## 12. Performance Considerations

- Indexes on foreign keys and date columns
- Partitioning by date for large fact tables
- Optimized join paths in star schema
- Incremental loading to reduce ETL time

## 13. Challenges & Solutions

During the development and implementation of the system, several challenges were encountered along with their corresponding solutions: Below these are those challenges and how we overcame them .

| Challenge | Solution |
| --- | --- |
| Inconsistent date formats | Standardized to ISO 8601 in ETL |
| Missing StudentID in room bookings | Flagged as "Walk-in" |
| Duplicate booking records | Removed using transaction ID deduplication |
| Department name variations | Mapped to standardized names |

**Table 3 : Showing Challenges & Solutions we faced**

## 14. Future Enhancements

- Real-time data streaming
- Predictive analytics for resource forecasting
- Integration with additional data sources (e.g., library events)
- Mobile-friendly dashboards
- Natural language query interface

## 15. Conclusion

The Library Usage Data Warehouse project successfully delivers a scalable, secure, and analytics-ready platform that transforms fragmented library data into actionable insights. By automating reporting, enhancing data quality, and enabling self-service analytics, the system empowers the University Library to make data-driven decisions, improve operational efficiency, and better serve its users.